Predicting with models

Predicting game outcome with Poisson regression

Poisson regression

First we are going to test the hypothesis that Home goals and Away goals are independent.

We will use a χ^2 test of independence of two categorical variables

$$H_0: Two\ variables\ are\ independent$$

$$H_1: Two\ variables\ are\ NOT\ independent$$

Calculate p-value and compare with significance level of α, if p-value < α nreject the NULL hypothesis

Poisson regression

First we are going to create categoric variables for Home and Away team goals in a way that more than 3 goals will be combined together

```
library(dplyr)
data("f_data_sm")
df <- f_data_sm %>% mutate(FTAG = ifelse(FTAG>=4, "4+", FTAG),
                           FTHG = ifelse(FTHG >=4, "4+", FTHG))
```

Poisson regression

recall that χ^2 test is using Expected and Observed frequencies as an input. Thus we will need to provide a table with joint frequencies of Home and away team goals

```
table(Away=df$FTAG, Home = df$FTHG)
```

```
##      Home
## Away     0     1     2     3    4+
##    0  7059  9325  7261  3998  2756
##    1  6074 10248  7644  4020  2621
##    2  3858  5243  4277  2092  1284
##    3  1688  2313  1464   826   462
##    4+ 1002  1103   742   300   154
```

Poisson regression

- We do the test for the last season
- we use function chisq.test
  - We specify table with Observed Frequencies
  - And specify simulate.p.values = TRUE: This will add robustness to our test
- We do the test for each country and extract p-values of the test

```
df %>% filter(SEASON==2019) %>% group_by(COUNTRY) %>%
  summarise(pvalue=chisq.test(table(FTAG, FTHG),
                              simulate.p.value=T)$p.value)
```

Poisson regression

```
df %>% filter(SEASON==2019) %>% group_by(COUNTRY) %>%
  summarise(pvalue=chisq.test(table(FTAG, FTHG),
                              simulate.p.value=T)$p.value)
```

```
## # A tibble: 11 x 2
##    COUNTRY      pvalue
##    <chr>         <dbl>
##  1 Belgium     0.241
##  2 England     0.00250
##  3 France      0.128
##  4 Germany     0.108
##  5 Greece      0.108
##  6 Italy       0.178
##  7 Netherlands 0.0245
##  8 Portugal    0.338
##  9 Scotland    0.222
## 10 Spain       0.858
## 11 Turkey      0.279
```

The hypothesis of independence is not rejected, at $\alpha = 0.05$, Thus they are independent

Poisson regression

- Poisson regression is used when the dependent variable is a count data, i.e. y=0,1,2,3…, thus we assume that the dependent variable has Poisson distribution
- Independent variables can be both numeric and categorical
- Poisson regression is a part of Generalized Linear Regression (GLM) models family

Poisson regression: the general definition

In this chapter we are fitting the model of the following form

$$\log(\lambda) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots \beta_2 X_2$$

Where λ is the mean of Poisson distribution and $\{X_1, \ldots, X_n\}$ are the independent variables

With the estimate of the λ we can generate Poisson distribution and calculate the probabilities for the number of the goals

Preparing data for the Poisson regression model

The initial data for Portugal, season 2018-2019

Making Predictions for Portugal Liga 1

```
data(f_data_sm)
portugal <- f_data_sm %>%
            filter(SEASON==2019, COUNTRY=="Portugal") %>%
            select(HOMETEAM, AWAYTEAM, FTHG, FTAG)

head(portugal, n=5)
```

```
##   HOMETEAM   AWAYTEAM FTHG FTAG
## 1  Benfica  Guimaraes    3    2
## 2    Porto     Chaves    5    0
## 3  Setubal       Aves    2    0
## 4  Tondela Belenenses    0    1
## 5 Feirense    Rio Ave    2    0
```

Preparing data for the Poisson regression model

The data need to be in the following format: 2 rows for each game and the indicator showing if the team is a home team

```
port1<-data.frame(portugal[,c("HOMETEAM", "AWAYTEAM", "FTHG")], Home=1)
port2<-data.frame(portugal[,c("AWAYTEAM", "HOMETEAM", "FTAG")], Home=0)
colnames(port1)<-c("Team", "Opponent", "Goal", "Home")
colnames(port2)<-c("Team", "Opponent","Goal", "Home")
portugal2<-rbind(port1, port2)
head(portugal2)
```

```
##       Team    Opponent Goal Home
## 1  Benfica   Guimaraes    3    1
## 2    Porto      Chaves    5    1
## 3  Setubal        Aves    2    1
## 4  Tondela  Belenenses    0    1
## 5 Feirense     Rio Ave    2    1
## 6 Maritimo Santa Clara    1    1
```

Preparing data for the Poisson regression model

Now lets have a look at the first game of the season: Benfica vs Guimaraes * Benfica was the home team, and the game ended with 3-2

```
portugal[1,]
```

```
##   HOMETEAM  AWAYTEAM FTHG FTAG
## 1  Benfica Guimaraes    3    2
```

- Lets look at our new dataset

```
portugal2[1,]
```

```
##      Team  Opponent Goal Home
## 1 Benfica Guimaraes    3    1
```

```
portugal2[307,]
```

```
##          Team Opponent Goal Home
## 307 Guimaraes  Benfica    2    0
```

The number of the rows in the portugal2 dataset is twice of the original one

```
dim(portugal)
```

```
## [1] 306   4
```

```
dim(portugal2)
```

```
## [1] 612   4
```

The model

```
model<-glm(Goal~Team+Opponent+Home, data=portugal2, family=poisson(link=log))
coefficients(model)
```

```
##          (Intercept)       TeamBelenenses          TeamBenfica
##          -0.04950176           0.18546812           1.06027038
##         TeamBoavista          TeamChaves         TeamFeirense
##          -0.04047050          -0.01881011          -0.24101272
##        TeamGuimaraes         TeamMaritimo       TeamMoreirense
##           0.25493167          -0.30424065           0.10213559
##         TeamNacional     TeamPortimonense           TeamPorto
##          -0.02792327           0.24259884           0.71365399
##          TeamRio Ave      TeamSanta Clara          TeamSetubal
##           0.36181966           0.20134445          -0.23625112
##         TeamSp Braga        TeamSp Lisbon          TeamTondela
##           0.45606737           0.70311363           0.14038603
##   OpponentBelenenses     OpponentBenfica     OpponentBoavista
##           0.04912813          -0.36998629          -0.20470680
##       OpponentChaves     OpponentFeirense    OpponentGuimaraes
##           0.15040090           0.25756387          -0.35243515
##     OpponentMaritimo   OpponentMoreirense     OpponentNacional
##          -0.11925272          -0.10282703           0.39741274
## OpponentPortimonense       OpponentPorto      OpponentRio Ave
##           0.19802795          -0.84856604           0.07904179
##  OpponentSanta Clara      OpponentSetubal     OpponentSp Braga
##          -0.07516828          -0.23759199          -0.25481347
##    OpponentSp Lisbon      OpponentTondela                 Home
##          -0.34879055           0.10390298           0.25808160
```

Making predictions

- Benfica is playing against Porto at home

First, we predict λ for Benfica

```
options(scipen=1, digits=4)
predict(model, data.frame(Home=1, Team="Benfica", Opponent="Porto"), type="response")
```

```
##     1
## 1.522
```

The same for Porto

```
options(scipen=1, digits=4)
predict(model, data.frame(Home=0, Team="Porto", Opponent="Benfica"),
        type="response")
```

```
##     1
## 1.342
```

When Benfica is playing Porto at Home, the expected (average) number of goals for Benfica is **1.522** and For Porto **1.342**

Lets do the calculations by hand for Benfica

Calculations of λ for Benfica

$$\log(\lambda) = \beta_0 + \beta_{TeamBenfica} + \beta_{OpponentPorto} + \beta_{Home}$$

```
options(scipen=1, digits=4)
coefficients(model)["(Intercept)"]
```

```
## (Intercept)
##     -0.0495
```

```
coefficients(model)["TeamBenfica"]
```

```
## TeamBenfica
##        1.06
```

```
coefficients(model)["OpponentPorto"]
```

```
## OpponentPorto
##       -0.8486
```

```
coefficients(model)["Home"]
```

```
##   Home
## 0.2581
```

Lets do the calculations by hand for Benfica

Calculations of λ for Benfica $\log(\lambda) = \beta_0 + \beta_{TeamBenfica} + \beta_{OpponentPorto} + \beta_{Home}$

```
options(scipen=1, digits=4)
coefficients(model)["(Intercept)"]
```

```
## (Intercept)
##     -0.0495
```

```
coefficients(model)["TeamBenfica"]
```

```
## TeamBenfica
##        1.06
```

```
coefficients(model)["OpponentPorto"]
```

```
## OpponentPorto
##       -0.8486
```

```
coefficients(model)["Home"]
```

```
##   Home
## 0.2581
```

$$\log(\lambda) = -0.0495 + 1.06 + (-0.8486) + 0.2581$$

$$\lambda = \exp(0.42\ ) = 1.52$$

Making predictions: Home advantage

The average expected goals of the home team divided by the average expected goals by the same team playing away will be equal to exponent of Home coefficient

```
# lambda for Benfica playing at home
b_h<-exp(-0.0495+1.06-0.8486 + 0.2581)
# lambda for Benfica playing against Porto away
b_a<-exp(-0.0495+1.06-0.8486)
b_h/b_a
```

```
## [1] 1.294
```

```
exp(coefficients(model)["Home"])
```

```
##  Home
## 1.294
```

Making predictions

As we have λ for both teams, we can use skellam distribution to come up with the probabilities

Simulate distribution for the goal difference in Benfica-Porto game

```
# Benfica's lambda
1.522
```

```
## [1] 1.522
```

```
# Porto's lambda
1.342
```

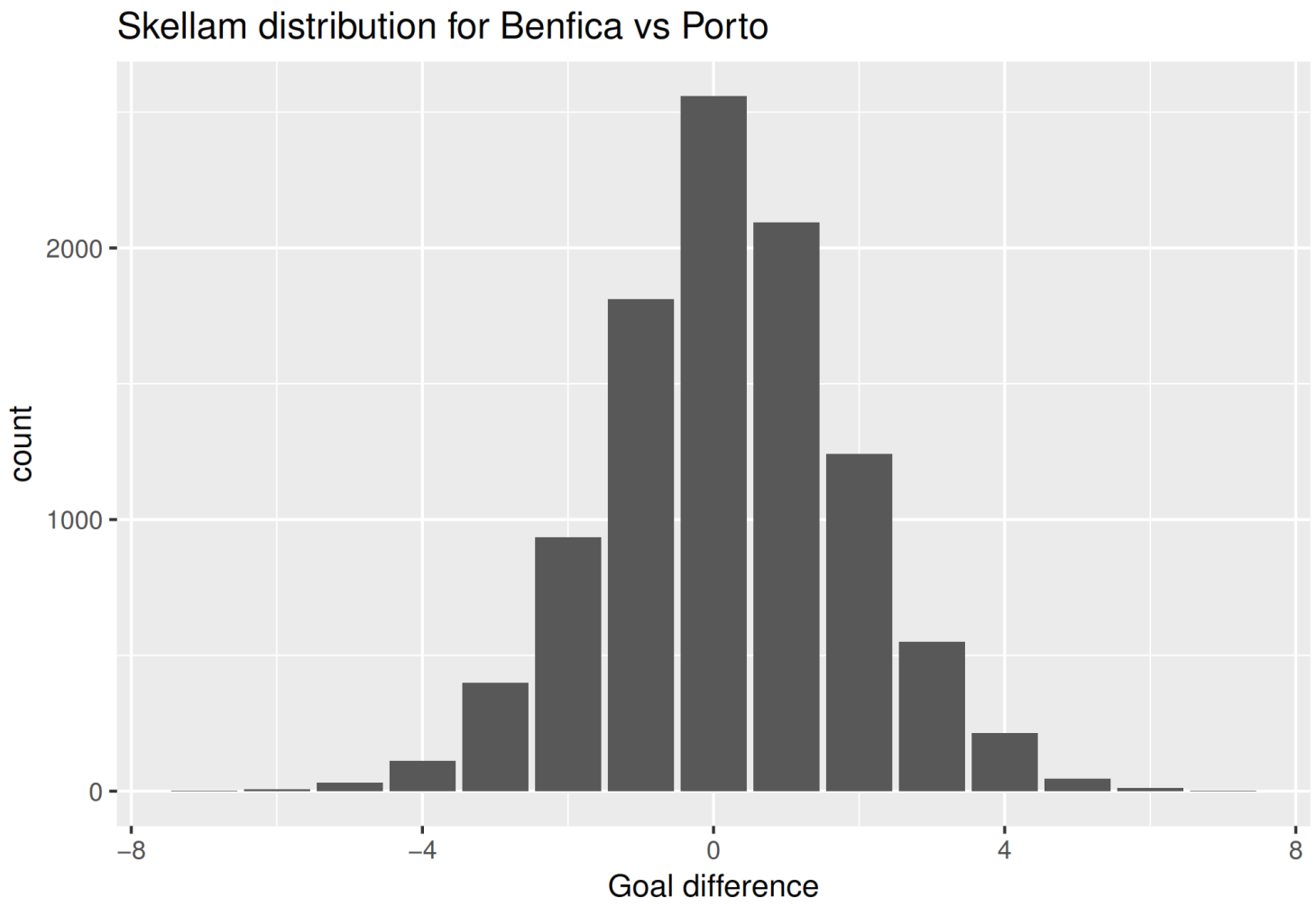
```
## [1] 1.342
```

```
# Simulate game results
set.seed(1)
k <- rskellam(10000,mu1 = 1.522, mu2 = 1.342)
```

Making predictions

A slight home team advantage

```
ggplot()+geom_bar(aes(x=k))+
  labs(x="Goal difference", title="Skellam distribution for Benfica vs Porto")
```



Predicting probabilities with Skellam distribution

Calculate the game output probabilities

Calculating the probabilities

```
# Away team to win
sum(dskellam(c(-100:-1),1.522, 1.342))
```

```
## [1] 0.3354
```

```
# Draw
sum(dskellam(0 ,1.522, 1.342))
```

```
## [1] 0.2484
```

```
# Home team to win
sum(dskellam(c(1:100),1.522, 1.342))
```

```
## [1] 0.4162
```

Bradley-Terry model

Bradley-Terry model

The Bradley-Terry model (Bradley and Terry 1952) assumes that in a 'contest' between any two 'players', say player i and player j ($i, j \in \{1, \ldots., K\}$) the ,odds that i beats j are α_i/α_j, where α_i and α_j are positive-valued parameters which represent the "ability" of each player/team

The general model definition

$$P(i\ beats\ j) = \frac{\alpha_i}{\alpha_i + \alpha_j}$$

Sometimes the modification with exponential score function

$$P(i\ beats\ j) = \frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}}$$

Bradley-Terry model

$$P(i\ beats\ j) = \frac{\alpha_i}{\alpha_i + \alpha_j}$$

$$logit\big(P(i\ beats\ j)\big) = \log\left(\frac{P(i\ beats\ j)}{P(j\ beats\ i)}\right) = \log\left(\frac{P(i\ beats\ j)}{1 - P(i\ beats\ j)}\right) =$$

$$\log\left(\frac{\alpha_i}{\alpha_i + \alpha_j} * \frac{\alpha_i + \alpha_j}{\alpha_j}\right) = \log(\alpha_i) - \log\big(\alpha_j\big) = \lambda_i - \lambda_j$$

The goal of the modeling is to find the best estimate for λ_i, λ_j

Bradley-Terry model

The modification with exponential score function

$$P(i\ beats\ j) = \frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}}$$

$$logit\big(P(i\ beats\ j)\big) = \log\left(\frac{P(i\ beats\ j)}{P(j\ beats\ i)}\right) = \log\left(\frac{P(i\ beats\ j)}{1 - P(i\ beats\ j)}\right) =$$

$$\log\left(\frac{e^{\alpha_i}}{e^{\alpha_i + e^{\alpha_j}}} * \frac{e^{\alpha_i + e^{\alpha_j}}}{e^{\alpha_j}}\right) = \alpha_i - \alpha_j$$

logit – log of odds

Bradley-Terry model

The basic Bradley-Terry model does not predict draw/tie games.

You can use *Davidson's* extension on Bradley-Terry model if want to predict draw:

https://www.jstor.org/stable/2283595?seq=1#page_scan_tab_contents

Bradley-Terry model

```
library(SportsAnalytics270)
data(nfl)
```

Look at the data

```
head(nfl, n=5)
```

```
##   schedule_season schedule_week schedule_date           team_away
## 1            2000             1    2000-09-03  San Francisco 49ers
## 2            2000             1    2000-09-03     Tennessee Titans
## 3            2000             1    2000-09-03 Jacksonville Jaguars
## 4            2000             1    2000-09-03  Philadelphia Eagles
## 5            2000             1    2000-09-03        New York Jets
##   score_away         team_home score_home             stadium
## 1         28   Atlanta Falcons         36        Georgia Dome
## 2         13     Buffalo Bills         16 Ralph Wilson Stadium
## 3         27  Cleveland Browns          7  FirstEnergy Stadium
## 4         41    Dallas Cowboys         14       Texas Stadium
## 5         20 Green Bay Packers         16       Lambeau Field
##   stadium_type stadium_location result spread home_odds_open
## 1       indoor      Atlanta, GA      H     -8             NA
## 2      outdoor Orchard Park, NY      H     -3             NA
## 3      outdoor    Cleveland, OH      A     20             NA
## 4      outdoor       Irving, TX      A     27             NA
## 5      outdoor    Green Bay, WI      A      4             NA
##   away_odds_open home_odds_close away_odds_close
## 1             NA              NA              NA
## 2             NA              NA              NA
## 3             NA              NA              NA
## 4             NA              NA              NA
## 5             NA              NA              NA
```

Bradley-Terry model

Data on NFL matches

Description
Format
A data frame with 4848 rows and 16 variables

schedule_season: from 2000-2016
schedule_week: The number of the week when the game was played
schedule_date: The actual date of the game
team_away: Away team
score_away: The score of the away team
team_home: Home team
score_home: The score of home team
stadium: The name of the stadium
stadium_type: indoor, outdoor, retractable
stadium_location: The location of the stadium: City and state abbreviation
odds: closing and opening betting odds
result: The result of the game: H=Home Win, T=Tie, A=Away Win
spread: points spread, the difference between away team and home team scores
home_odds_open, home_odds_close: Opening and closing decimal odds for the home team
away_odds_open, away_odds_open: Opening and closing decimal odds for the away team

Bradley-Terry model

Some data preparation

```
nfl2 <- nfl%>%
  select(team_home, team_away, result) %>%
  filter(result !="T") %>%
  mutate(ht_w = ifelse(result=="H",1,0),
         at_w = ifelse(result=="A",1,0))
head(nfl2)
```

```
##            team_home            team_away result ht_w at_w
## 1    Atlanta Falcons  San Francisco 49ers      H    1    0
## 2      Buffalo Bills     Tennessee Titans      H    1    0
## 3   Cleveland Browns Jacksonville Jaguars      A    0    1
## 4     Dallas Cowboys  Philadelphia Eagles      A    0    1
## 5  Green Bay Packers        New York Jets      A    0    1
## 6 Kansas City Chiefs   Indianapolis Colts      A    0    1
```

Bradley-Terry model

The final dataset

```
library(dplyr)
nfl3<-nfl2 %>%
    mutate(team_home=as.factor(team_home),
           team_away = as.factor(team_away)) %>%
    group_by(team_home, team_away) %>%
    summarise(ht=sum(ht_w), at=sum(at_w))

head(nfl3, n=5)
```

```
## # A tibble: 5 x 4
## # Groups:   team_home [1]
##   team_home         team_away            ht    at
##   <fct>             <fct>             <dbl> <dbl>
## 1 Arizona Cardinals Atlanta Falcons       2     1
## 2 Arizona Cardinals Baltimore Ravens      1     2
## 3 Arizona Cardinals Buffalo Bills         1     1
## 4 Arizona Cardinals Carolina Panthers     2     4
## 5 Arizona Cardinals Chicago Bears         0     2
```

Bradley-Terry model

```
library(BradleyTerry2)
model<-BTm(cbind(ht, at), team_home, team_away, data=nfl3, id="team_")
```

Bradley-Terry model

```
summary(model)
```

```
##
## Call:
## BTm(outcome = cbind(ht, at), player1 = team_home, player2 = team_away,
##     id = "team_", data = nfl3)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -2.7364  -0.2698   0.2822   1.2011   3.2514
##
## Coefficients:
##                           Estimate Std. Error z value Pr(>|z|)
## team_Atlanta Falcons       0.36746    0.16447   2.234 0.025472 *
## team_Baltimore Ravens      0.61137    0.16993   3.598 0.000321 ***
## team_Buffalo Bills         0.05251    0.16990   0.309 0.757269
## team_Carolina Panthers     0.30602    0.16391   1.867 0.061902 .
## team_Chicago Bears         0.24153    0.16537   1.461 0.144142
## team_Cincinnati Bengals    0.23362    0.16957   1.378 0.168285
## team_Cleveland Browns     -0.42056    0.17443  -2.411 0.015903 *
## team_Dallas Cowboys        0.41274    0.16354   2.524 0.011608 *
## team_Denver Broncos        0.61867    0.16867   3.668 0.000244 ***
## team_Detroit Lions        -0.22572    0.16623  -1.358 0.174506
## team_Green Bay Packers     0.68295    0.16739   4.080 4.50e-05 ***
## team_Houston Texans        0.08908    0.17370   0.513 0.608075
## team_Indianapolis Colts    0.80328    0.17108   4.695 2.66e-06 ***
## team_Jacksonville Jaguars -0.08200    0.16969  -0.483 0.628948
## team_Kansas City Chiefs    0.33316    0.16759   1.988 0.046826 *
## team_Los Angeles Chargers  0.36800    0.16761   2.196 0.028127 *
## team_Los Angeles Rams      0.04914    0.15655   0.314 0.753615
## team_Miami Dolphins        0.28758    0.16917   1.700 0.089139 .
## team_Minnesota Vikings     0.32602    0.16472   1.979 0.047790 *
## team_New England Patriots  1.34835    0.17977   7.501 6.36e-14 ***
## team_New Orleans Saints    0.50572    0.16547   3.056 0.002241 **
## team_New York Giants       0.32258    0.16354   1.973 0.048552 *
## team_New York Jets         0.22661    0.16940   1.338 0.180998
## team_Oakland Raiders      -0.10695    0.16917  -0.632 0.527270
## team_Philadelphia Eagles   0.65415    0.16495   3.966 7.32e-05 ***
## team_Pittsburgh Steelers   0.90424    0.17328   5.218 1.81e-07 ***
## team_San Francisco 49ers   0.05031    0.15632   0.322 0.747575
## team_Seattle Seahawks      0.47580    0.15737   3.024 0.002499 **
## team_Tampa Bay Buccaneers  0.04106    0.16579   0.248 0.804414
## team_Tennessee Titans      0.28242    0.16862   1.675 0.093953 .
## team_Washington Redskins   0.03768    0.16408   0.230 0.818354
##
```

Coefficients represent the teams abilities to win

Bradley-Terry model

```
coef<-model$coefficients
sort(coef, decreasing = T)
```

```
## team_New England Patriots  team_Pittsburgh Steelers
##                 1.34835210                0.90423628
##   team_Indianapolis Colts    team_Green Bay Packers
##                 0.80328130                0.68294804
##  team_Philadelphia Eagles       team_Denver Broncos
##                 0.65415054                0.61866889
##     team_Baltimore Ravens   team_New Orleans Saints
##                 0.61137135                0.50571851
##     team_Seattle Seahawks       team_Dallas Cowboys
##                 0.47579994                0.41274035
## team_Los Angeles Chargers      team_Atlanta Falcons
##                 0.36799652                0.36746020
##   team_Kansas City Chiefs    team_Minnesota Vikings
##                 0.33315530                0.32602364
##      team_New York Giants    team_Carolina Panthers
##                 0.32257683                0.30602400
##       team_Miami Dolphins     team_Tennessee Titans
##                 0.28758429                0.28242436
##        team_Chicago Bears   team_Cincinnati Bengals
##                 0.24153047                0.23362130
##        team_New York Jets       team_Houston Texans
##                 0.22661028                0.08907555
##        team_Buffalo Bills  team_San Francisco 49ers
##                 0.05251068                0.05030864
##     team_Los Angeles Rams team_Tampa Bay Buccaneers
##                 0.04913867                0.04105585
##  team_Washington Redskins team_Jacksonville Jaguars
##                 0.03768352               -0.08199798
##      team_Oakland Raiders        team_Detroit Lions
##                -0.10694660               -0.22571521
##     team_Cleveland Browns
##                -0.42056234
```



New England Patriots: the strongest team
Cleveland Browns: the weakest team



Bradley-Terry model

Overall there are 32 teams in NFL

```
length(unique(nfl3$team_home))
```

```
## [1] 32
```

We have coefficients for 31 teams

```
length(coef)
```

```
## [1] 31
```

- One team is taken as a reference level
- Its ability is = 0
- In our case that is Arizona Cardinals as it comes first alphabetically

Bradley-Terry model

```
BTabilities(model)
```

```
##                          ability      s.e.
## Arizona Cardinals     0.00000000 0.0000000
## Atlanta Falcons       0.36746020 0.1644732
## Baltimore Ravens      0.61137135 0.1699258
## Buffalo Bills         0.05251068 0.1698996
## Carolina Panthers     0.30602400 0.1639125
## Chicago Bears         0.24153047 0.1653708
## Cincinnati Bengals    0.23362130 0.1695692
## Cleveland Browns     -0.42056234 0.1744254
## Dallas Cowboys        0.41274035 0.1635357
## Denver Broncos        0.61866889 0.1686659
## Detroit Lions        -0.22571521 0.1662277
## Green Bay Packers     0.68294804 0.1673872
## Houston Texans        0.08907555 0.1736962
## Indianapolis Colts    0.80328130 0.1710844
## Jacksonville Jaguars -0.08199798 0.1696949
## Kansas City Chiefs    0.33315530 0.1675943
## Los Angeles Chargers  0.36799652 0.1676134
## Los Angeles Rams      0.04913867 0.1565545
## Miami Dolphins        0.28758429 0.1691714
## Minnesota Vikings     0.32602364 0.1647226
## New England Patriots  1.34835210 0.1797679
## New Orleans Saints    0.50571851 0.1654709
## New York Giants       0.32257683 0.1635367
## New York Jets         0.22661028 0.1694044
## Oakland Raiders      -0.10694660 0.1691713
## Philadelphia Eagles   0.65415054 0.1649516
## Pittsburgh Steelers   0.90423628 0.1732802
## San Francisco 49ers   0.05030864 0.1563161
## Seattle Seahawks      0.47579994 0.1573667
## Tampa Bay Buccaneers  0.04105585 0.1657889
## Tennessee Titans      0.28242436 0.1686211
## Washington Redskins   0.03768352 0.1640823
```

- Abilities are calculated relative to Arizona Cardinales
- Thus, **positive** ability means that the team is doing better than Cardinals
  - Negative ability is showing that the team is doing worst than Cardinals

Bradley-Terry model

```
abilities<-as.data.frame(BTabilities(model))
abilities$team <- rownames(abilities)
abilities<-abilities[order(abilities$ability, decreasing = T),]
abilities
```

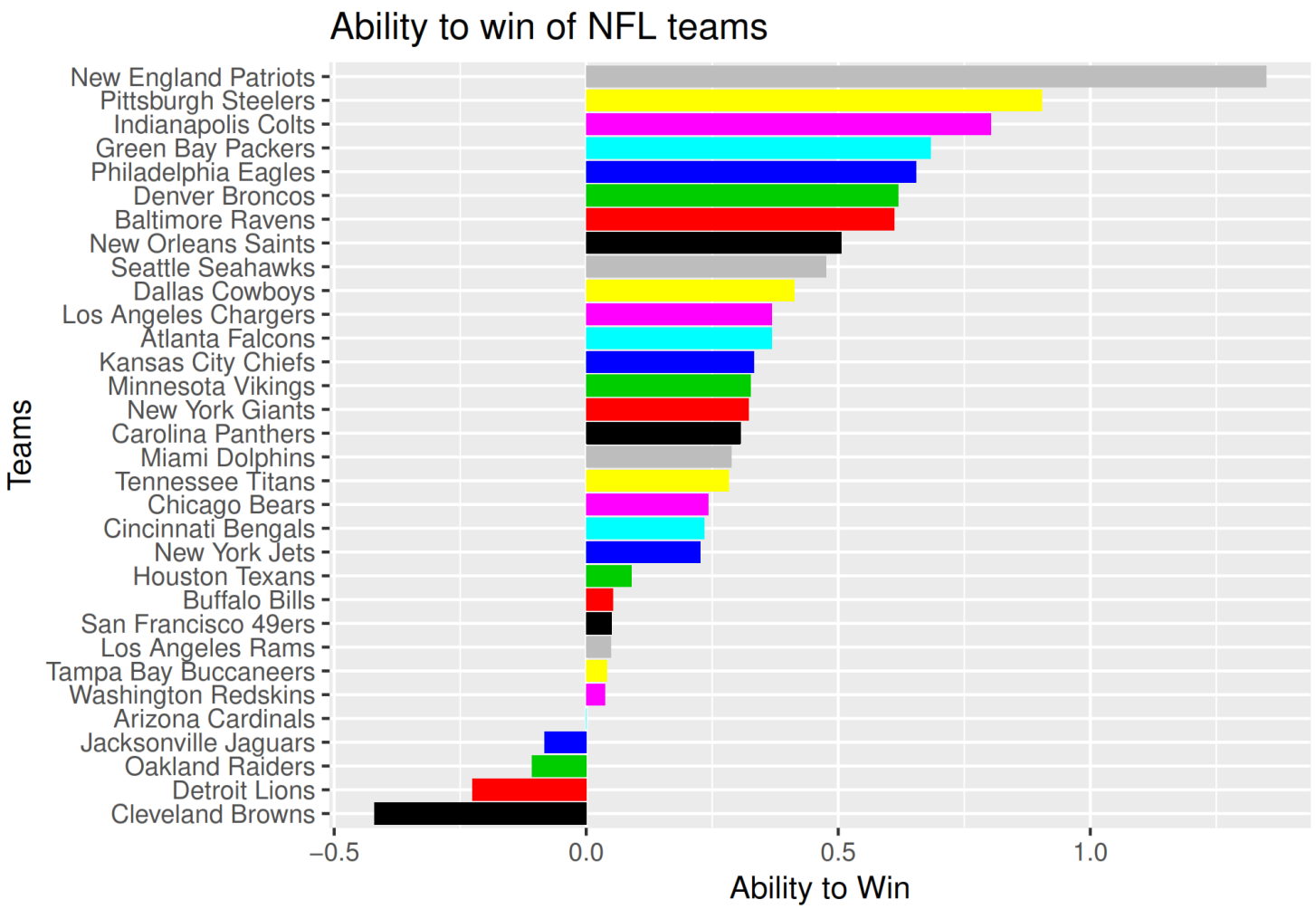
```
##                          ability      s.e.                  team
## New England Patriots  1.34835210 0.1797679 New England Patriots
## Pittsburgh Steelers   0.90423628 0.1732802  Pittsburgh Steelers
## Indianapolis Colts    0.80328130 0.1710844   Indianapolis Colts
## Green Bay Packers     0.68294804 0.1673872    Green Bay Packers
## Philadelphia Eagles   0.65415054 0.1649516  Philadelphia Eagles
## Denver Broncos        0.61866889 0.1686659       Denver Broncos
## Baltimore Ravens      0.61137135 0.1699258     Baltimore Ravens
## New Orleans Saints    0.50571851 0.1654709   New Orleans Saints
## Seattle Seahawks      0.47579994 0.1573667     Seattle Seahawks
## Dallas Cowboys        0.41274035 0.1635357       Dallas Cowboys
## Los Angeles Chargers  0.36799652 0.1676134 Los Angeles Chargers
## Atlanta Falcons       0.36746020 0.1644732      Atlanta Falcons
## Kansas City Chiefs    0.33315530 0.1675943   Kansas City Chiefs
## Minnesota Vikings     0.32602364 0.1647226    Minnesota Vikings
## New York Giants       0.32257683 0.1635367      New York Giants
## Carolina Panthers     0.30602400 0.1639125    Carolina Panthers
## Miami Dolphins        0.28758429 0.1691714       Miami Dolphins
## Tennessee Titans      0.28242436 0.1686211     Tennessee Titans
## Chicago Bears         0.24153047 0.1653708        Chicago Bears
## Cincinnati Bengals    0.23362130 0.1695692   Cincinnati Bengals
## New York Jets         0.22661028 0.1694044        New York Jets
## Houston Texans        0.08907555 0.1736962       Houston Texans
## Buffalo Bills         0.05251068 0.1698996        Buffalo Bills
## San Francisco 49ers   0.05030864 0.1563161  San Francisco 49ers
## Los Angeles Rams      0.04913867 0.1565545     Los Angeles Rams
## Tampa Bay Buccaneers  0.04105585 0.1657889 Tampa Bay Buccaneers
## Washington Redskins   0.03768352 0.1640823  Washington Redskins
## Arizona Cardinals     0.00000000 0.0000000    Arizona Cardinals
## Jacksonville Jaguars -0.08199798 0.1696949 Jacksonville Jaguars
## Oakland Raiders      -0.10694660 0.1691713      Oakland Raiders
## Detroit Lions        -0.22571521 0.1662277        Detroit Lions
## Cleveland Browns     -0.42056234 0.1744254     Cleveland Browns
```

- Let’s do some plotting
- First let's create a dataframe with abilities

Bradley-Terry model

Abilities from strongest to weakest

```
ggplot(data=abilities, aes(x=reorder(team,ability), y=ability)) +
  geom_bar(stat="identity", fill=1:32) + coord_flip()+
  labs(x="Teams", y="Ability to Win", title="Ability to win of NFL teams")
```



Bradley-Terry model

- Lets say we want to calculate the abilities relative to the weakest team, so all the abilities would be ≥ 0
- You could do this by releveling the team variables

```
nfl4<-nfl2 %>%
    mutate(team_home = relevel(team_home, ref="Cleveland Browns"),
           team_away = relevel(team_away, ref="Cleveland Browns")) %>%
    group_by(team_home, team_away) %>%
    summarise(ht=sum(ht_w), at=sum(at_w))

model_browns <- BTm(cbind(ht, at), team_home, team_away, data=nfl4, id="team_")
```

OR

Standardize the variable ability around its minimum (these two approaches will give exactly the same result)

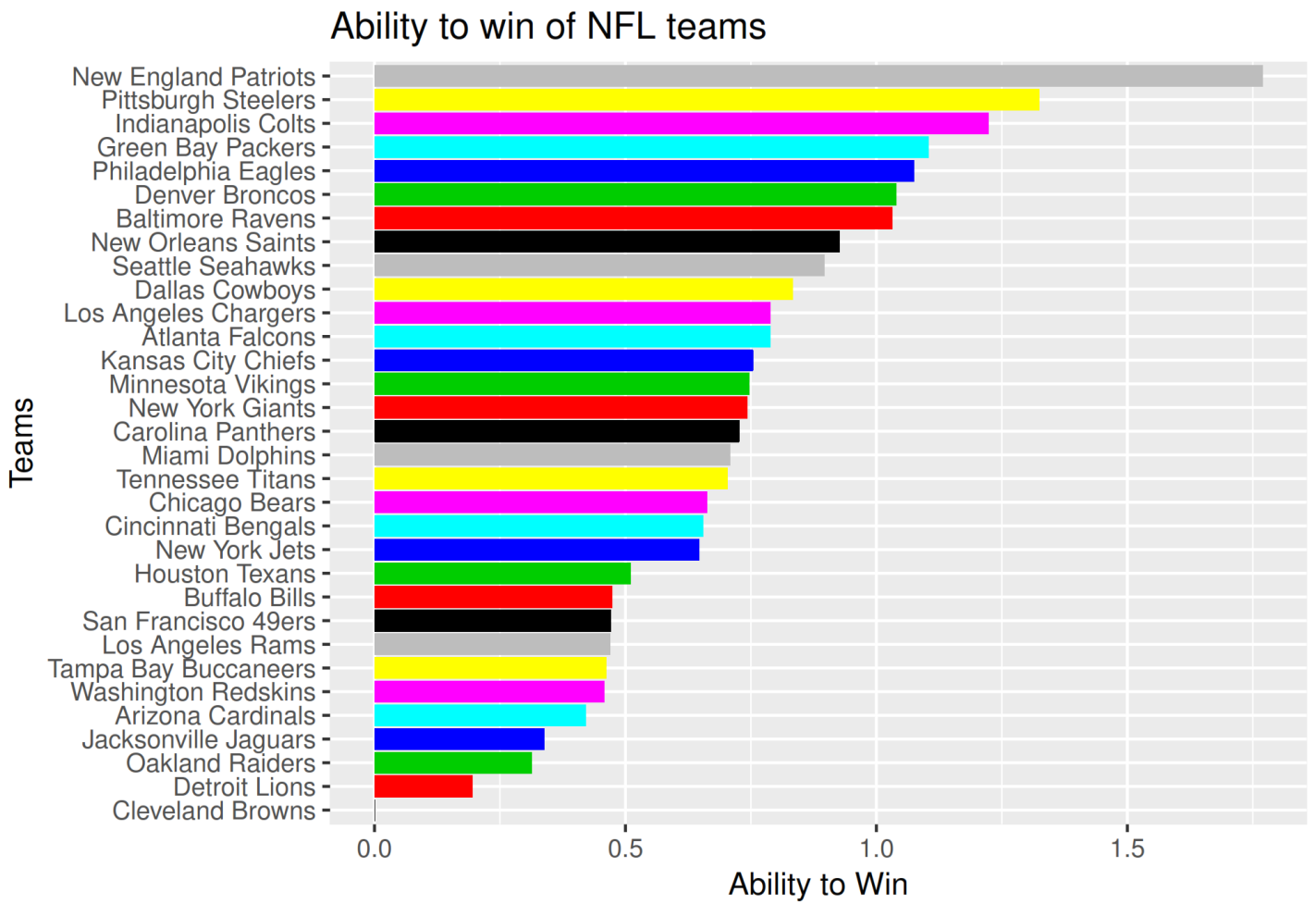
```
abilities_brown <- abilities
abilities_brown$ability <- abilities_brown$ability - min(abilities$ability)
head(abilities_brown)
```

```
##                         ability      s.e.                  team
## New England Patriots 1.768914 0.1797679 New England Patriots
## Pittsburgh Steelers  1.324799 0.1732802  Pittsburgh Steelers
## Indianapolis Colts   1.223844 0.1710844   Indianapolis Colts
## Green Bay Packers    1.103510 0.1673872    Green Bay Packers
## Philadelphia Eagles  1.074713 0.1649516  Philadelphia Eagles
## Denver Broncos       1.039231 0.1686659       Denver Broncos
```

Bradley-Terry model

Abilities plot

```
ggplot(data=abilities_brown, aes(x=reorder(team,ability), y=ability)) +
  geom_bar(stat="identity", fill=1:32) + coord_flip()+
  labs(x="Teams", y="Ability to Win", title="Ability to win of NFL teams")
```



Bradley-Terry model

Lets do predictions for a sample of games for Denver Broncos

```
broncos<-data.frame(team_home=rep("Denver Broncos", 4),
                    team_away=c("Cleveland Browns", "Tennessee Titans",
                                "Chicago Bears", "Detroit Lions"))
# we need to make team_home and team_away having the same factor structure
# as in the initial model

broncos$team_home<-factor(broncos$team_home,
                          levels(nfl3$team_away))

broncos$team_away<-factor(broncos$team_away,
                          levels(nfl3$team_away))
str(broncos)
```

```
## 'data.frame':    4 obs. of  2 variables:
##  $ team_home: Factor w/ 32 levels "Arizona Cardinals",..: 10 10 10 10
##  $ team_away: Factor w/ 32 levels "Arizona Cardinals",..: 8 31 6 11
```

```
broncos
```

```
##        team_home        team_away
## 1 Denver Broncos Cleveland Browns
## 2 Denver Broncos Tennessee Titans
## 3 Denver Broncos    Chicago Bears
## 4 Denver Broncos    Detroit Lions
```



Bradley-Terry model

```
#This is the probability of Denver Broncos winning

broncos_prob<-predict(model, newdata=broncos, level=2, type="response")
broncos_prob
```

```
##         1         2         3         4
## 0.7387016 0.5832780 0.5931827 0.6993878
```

Make a dataframe with probs

```
broncos_df<-data.frame(broncos, ht_w=broncos_prob,
                                at_w=1-broncos_prob)
broncos_df
```

```
##        team_home        team_away      ht_w      at_w
## 1 Denver Broncos Cleveland Browns 0.7387016 0.2612984
## 2 Denver Broncos Tennessee Titans 0.5832780 0.4167220
## 3 Denver Broncos    Chicago Bears 0.5931827 0.4068173
## 4 Denver Broncos    Detroit Lions 0.6993878 0.3006122
```

Bradley-Terry model

Probabilities are predicted using the following formula

$$P(i\ beats\ j) = \frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}}$$

$$P(j\ beats\ i) = \frac{e^{\alpha_j}}{e^{\alpha_i} + e^{\alpha_j}}$$

Bradley-Terry model

Calculating probabilities manually for Denver Broncos-Cleveland Browns

```
#first get abbilities
broncos_ab<-abilities[rownames(abilities)=="Denver Broncos",1]
browns_ab<-abilities[rownames(abilities)=="Cleveland Browns",1]

broncos_ab
```

```
## [1] 0.6186689
```

```
browns_ab
```

```
## [1] -0.4205623
```

```
# probability of Broncos to win
exp(broncos_ab)/(exp(broncos_ab)+exp(browns_ab))
```

```
## [1] 0.7387016
```

$$P(i\ beats\ j) = \frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}}$$

Bradley-Terry model with order effect

Bradley-Terry model with order effect

- In certain types of application some or all contests have an associated `bias', related to the order in which items are presented to a judge or with the location in which a contest takes place, for example.

- In sports this can be related to home advantage: the ability of the team is different when it plays at home compared to when it plays away.

- In this case the model has the following structure

$$logit\big(P(i\ beats\ j)\big) = \alpha_i - \alpha_j + \delta z$$

$z = 1$ if i has the supposed advantage and
$z = -1$ if j has it
$z = 0$ if there is no advantage

Bradley-Terry model with order effect

We need to do several data transformations to get the model to account home advantage factor.

```
nfl3$team_home <- data.frame(team = nfl3$team_home, at.home = 1)
nfl3$team_away <- data.frame(team = nfl3$team_away, at.home = 0)
```

The result is a grouped_df

```
str(nfl3)
```

```
## Classes 'grouped_df', 'tbl_df', 'tbl' and 'data.frame':  992 obs. of  4 variables:
##  $ team_home:'data.frame':   992 obs. of  2 variables:
##   ..$ team   : Factor w/ 32 levels "Arizona Cardinals",..: 1 1 1 1 1 1 1 1 1 1 ...
##   ..$ at.home: num  1 1 1 1 1 1 1 1 1 1 ...
##  $ team_away:'data.frame':   992 obs. of  2 variables:
##   ..$ team   : Factor w/ 32 levels "Arizona Cardinals",..: 2 3 4 5 6 7 8 9 10 11 ...
##   ..$ at.home: num  0 0 0 0 0 0 0 0 0 0 ...
##  $ ht        : num  2 1 1 2 0 2 3 6 1 7 ...
##  $ at        : num  1 2 1 4 2 0 0 2 2 0 ...
##  - attr(*, "vars")= chr "team_home"
##  - attr(*, "drop")= logi TRUE
```

Bradley-Terry model with order effect

```
head(nfl3$team_home, n=5)
```

```
##                  team at.home
## 1 Arizona Cardinals       1
## 2 Arizona Cardinals       1
## 3 Arizona Cardinals       1
## 4 Arizona Cardinals       1
## 5 Arizona Cardinals       1
```

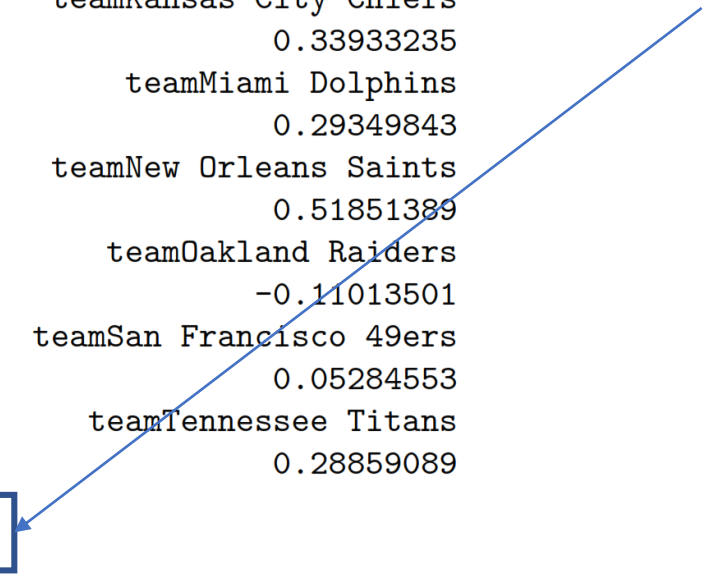
Bradley-Terry model with order effect

```
model2<-BTm(cbind(ht, at), team_home, team_away, formula=~team+at.home,
            data=nfl3, id="team")
```

```
model2$coefficients
```

```
##      teamAtlanta Falcons     teamBaltimore Ravens       teamBuffalo Bills
##                0.37386375               0.62493459               0.05366597
##    teamCarolina Panthers       teamChicago Bears   teamCincinnati Bengals
##                0.31116634               0.24678148               0.24130479
##     teamCleveland Browns       teamDallas Cowboys      teamDenver Broncos
##               -0.42821232               0.42081176               0.63229784
##        teamDetroit Lions    teamGreen Bay Packers      teamHouston Texans
##               -0.23141857               0.69886104               0.09026614
##   teamIndianapolis Colts teamJacksonville Jaguars   teamKansas City Chiefs
##                0.82082061              -0.08398155               0.33933235
## teamLos Angeles Chargers     teamLos Angeles Rams      teamMiami Dolphins
##                0.37466974               0.04644809               0.29349843
##    teamMinnesota Vikings teamNew England Patriots   teamNew Orleans Saints
##                0.32999963               1.37599428               0.51851389
##      teamNew York Giants        teamNew York Jets     teamOakland Raiders
##                0.32672185               0.23093403              -0.11013501
##  teamPhiladelphia Eagles  teamPittsburgh Steelers teamSan Francisco 49ers
##                0.66637647               0.92438241               0.05284553
##     teamSeattle Seahawks teamTampa Bay Buccaneers    teamTennessee Titans
##                0.48432827               0.04148136               0.28859089
##  teamWashington Redskins                  at.home
##                0.03613749               0.30284657
```

The home team advantage



Bradley-Terry model with order effect

Let make predictions for Broncos vs Browns by hand, assuming that Broncos are playing at home

```
db <- coefficients(model2)['teamDenver Broncos']
db
```

```
## teamDenver Broncos
##          0.6322978
```

```
cb <- coefficients(model2)['teamCleveland Browns']
cb
```

```
## teamCleveland Browns
##           -0.4282123
```

```
h <- coefficients(model2)['at.home']
h
```

```
##   at.home
## 0.3028466
```

Bradley-Terry model with order effect

$$P(i \text{ beats } j) = \frac{e^{\alpha_i + h}}{e^{\alpha_i + h} + e^{\alpha_j}}$$

Probability of Broncos to win at home

```
w_h <- exp(db+h)/(exp(db+h)+exp(cb))
w_h
```

```
## teamDenver Broncos
##          0.7963047
```

Odds of Broncos to win at home

```
od_h <- w_h/(1-w_h)
od_h
```

```
## teamDenver Broncos
##          3.909294
```

Bradley-Terry model with order effect

Playing at “neutral” field, when no one has advantage (z=0)

$$P(i\ beats\ j) = \frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}}$$

```
w_n <- exp(db)/(exp(db)+exp(cb))
w_n
```

```
## teamDenver Broncos
##          0.742788
```

Odds of wining at “neutral” field

```
od_n <- w_n/(1-w_n)
od_n
```

```
## teamDenver Broncos
##          2.887844
```

Bradley-Terry model with order effect

Odds ratio and home team advantage

```
od_h/od_n
```

```
## teamDenver Broncos
##           1.353707
```

```
exp(coefficients(model2)['at.home'])
```

```
##  at.home
## 1.353707
```

$e^{at.home}$ is the odd multiplier for a home team to win compared to the neutral field

Bradley-Terry model with order effect

The same game, now Broncos is away, Browns is at home

$$P(j \text{ beats } i) = \frac{e^{\alpha_j + h}}{e^{\alpha_i} + e^{\alpha_j + h}}$$

```
w_a <- exp(db-h)/(exp(db-h)+exp(cb))
w_a
```

```
## teamDenver Broncos
##          0.6808463
```

Probability of Browns to win at home

```
1 - exp(db-h)/(exp(db-h)+exp(cb))
```

```
## teamDenver Broncos
##          0.3191537
```

```
# OR
exp(cb+h)/(exp(db)+exp(cb+h))
```

```
## teamCleveland Browns
##           0.3191537
```

Bradley-Terry model with order effect

Making predictions for the series of games

```
broncos<-data.frame(team_home=rep("Denver Broncos", 4),
                    team_away=c("Cleveland Browns", "Tennessee Titans",
                                "Chicago Bears", "Detroit Lions"))

broncos$team_home<-data.frame(team = factor(broncos$team_home,
                                        levels(nfl3$team_home$team)),   at.home = 1)
broncos$team_away<-data.frame(team = factor(broncos$team_away,
                                              levels(nfl3$team_home$team)),at.home = 0)
broncos
```

```
##   team_home.team team_home.at.home   team_away.team team_away.at.home
## 1 Denver Broncos                 1 Cleveland Browns                 0
## 2 Denver Broncos                 1 Tennessee Titans                 0
## 3 Denver Broncos                 1    Chicago Bears                 0
## 4 Denver Broncos                 1    Detroit Lions                 0
```

Bradley-Terry model with order effect

```
br2 <- predict(model2, newdata=broncos, level=2, type="response")
br2
```

```
##         1         2         3         4
## 0.7963047 0.6562334 0.6656027 0.7625232
```